Contrast input and manual interventions significantly affect FreeSurfer morphometry and clinical correlations

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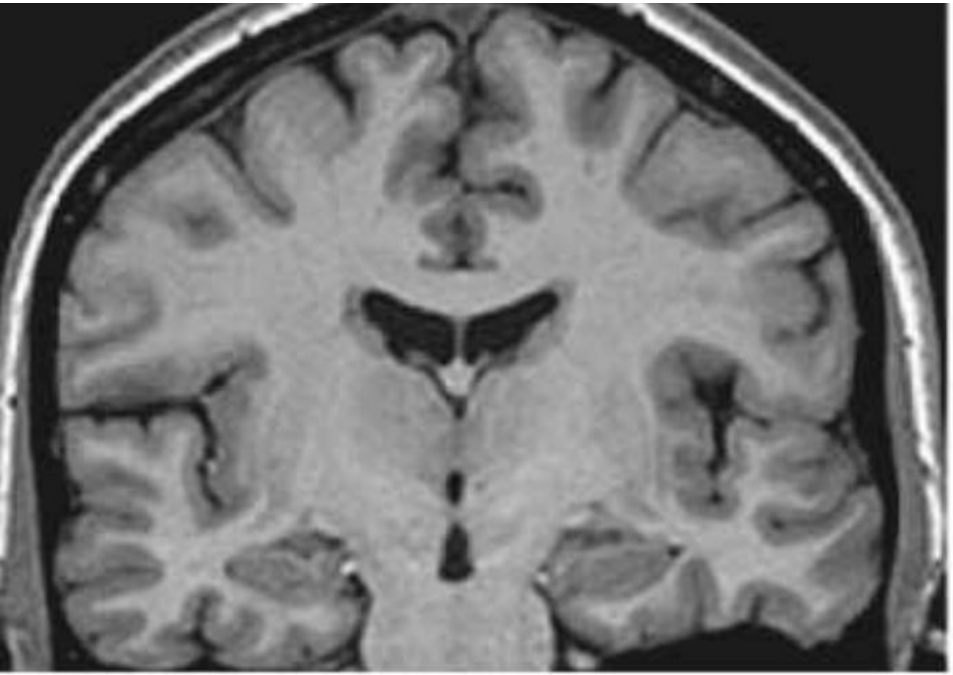
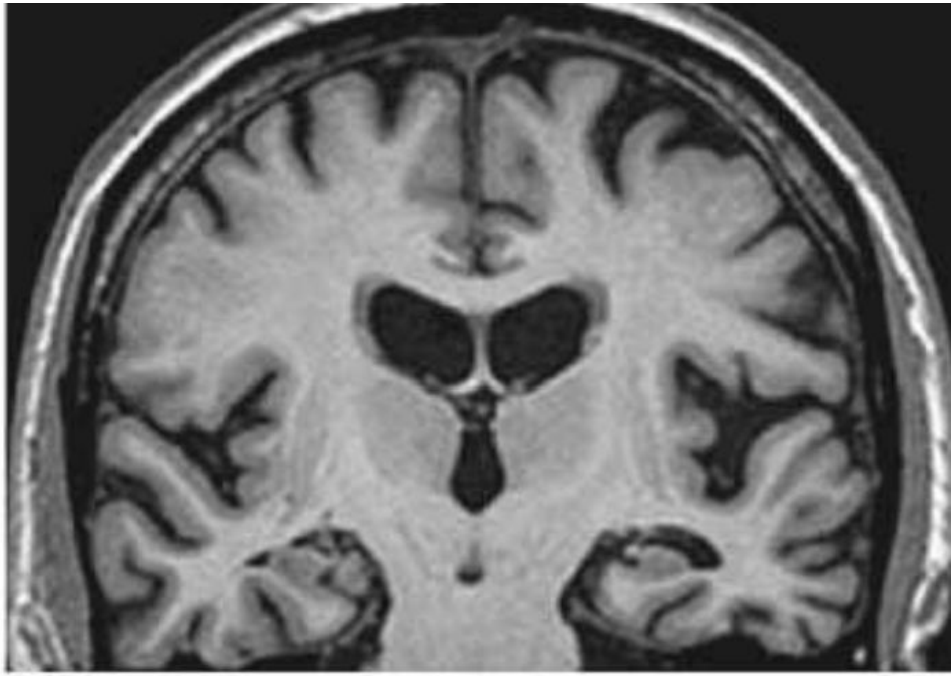
Context: Study cases



Healthy

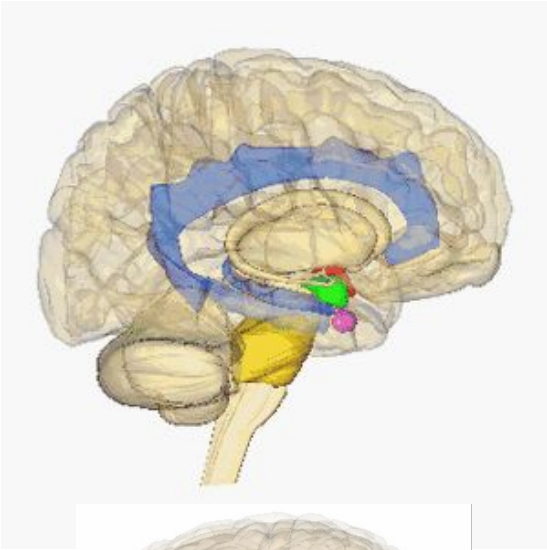


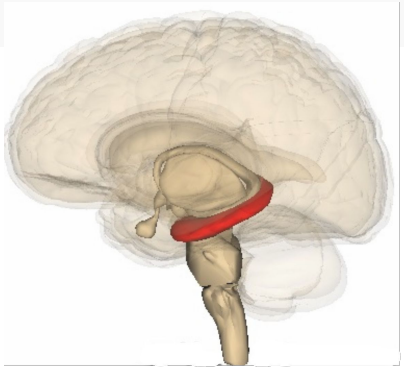
Alzheimer's

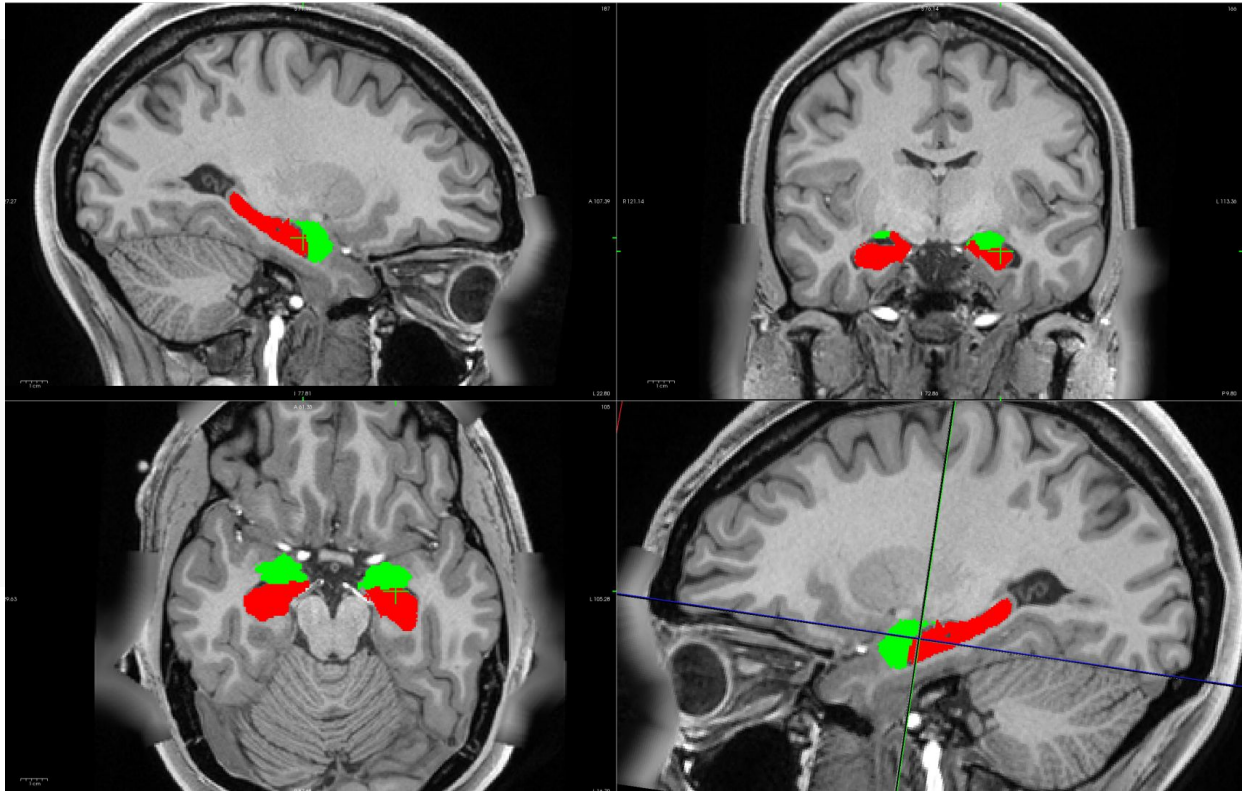
Freesurfer pipeline 'recon-all' used for quantification of **brain morphometrics (?)** in studies of **neurodegeneration and aging**

Context: Morphometric measures of interest

Hippocampal volume



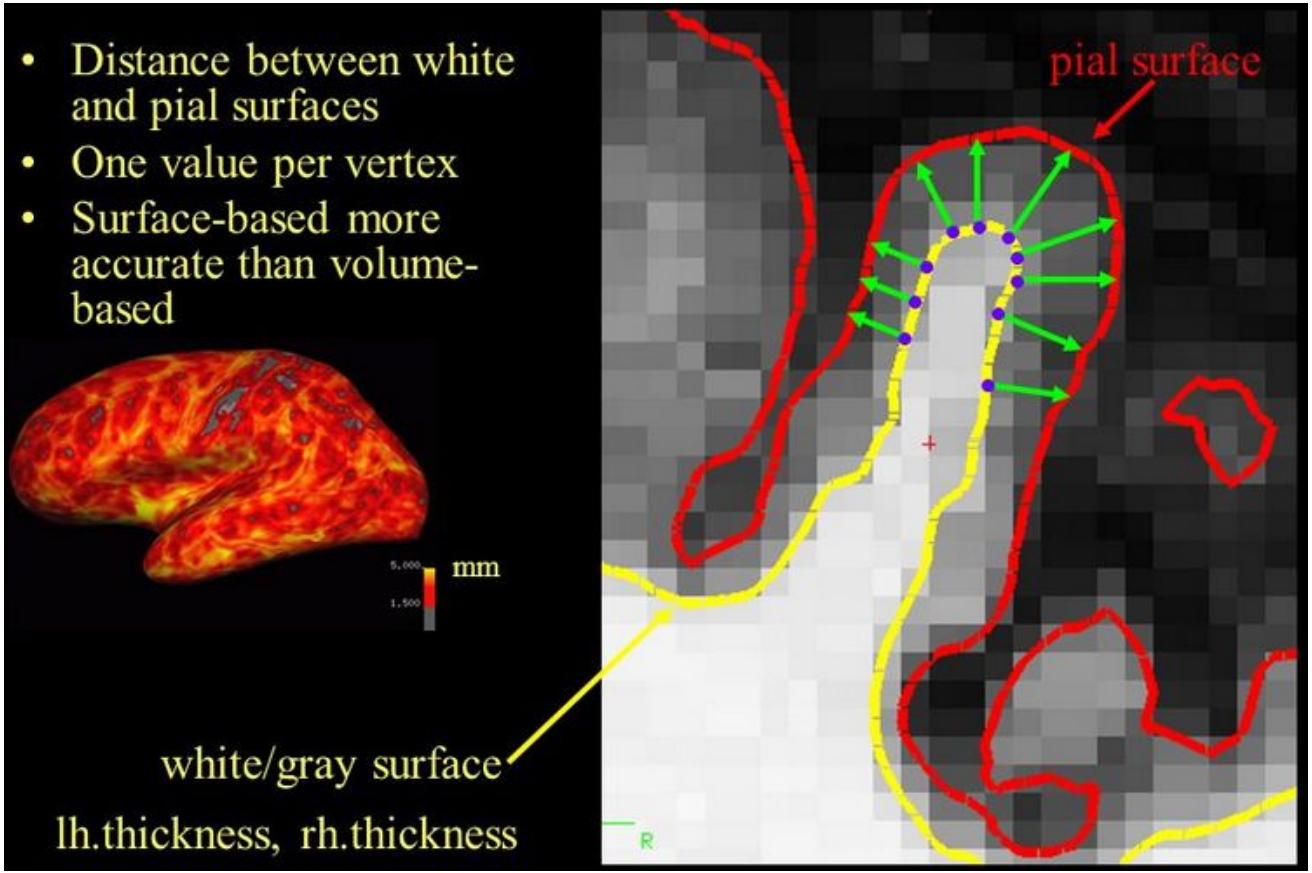




Hippocampus (# voxels or mm^3)

Context: Morphometric measures of interest

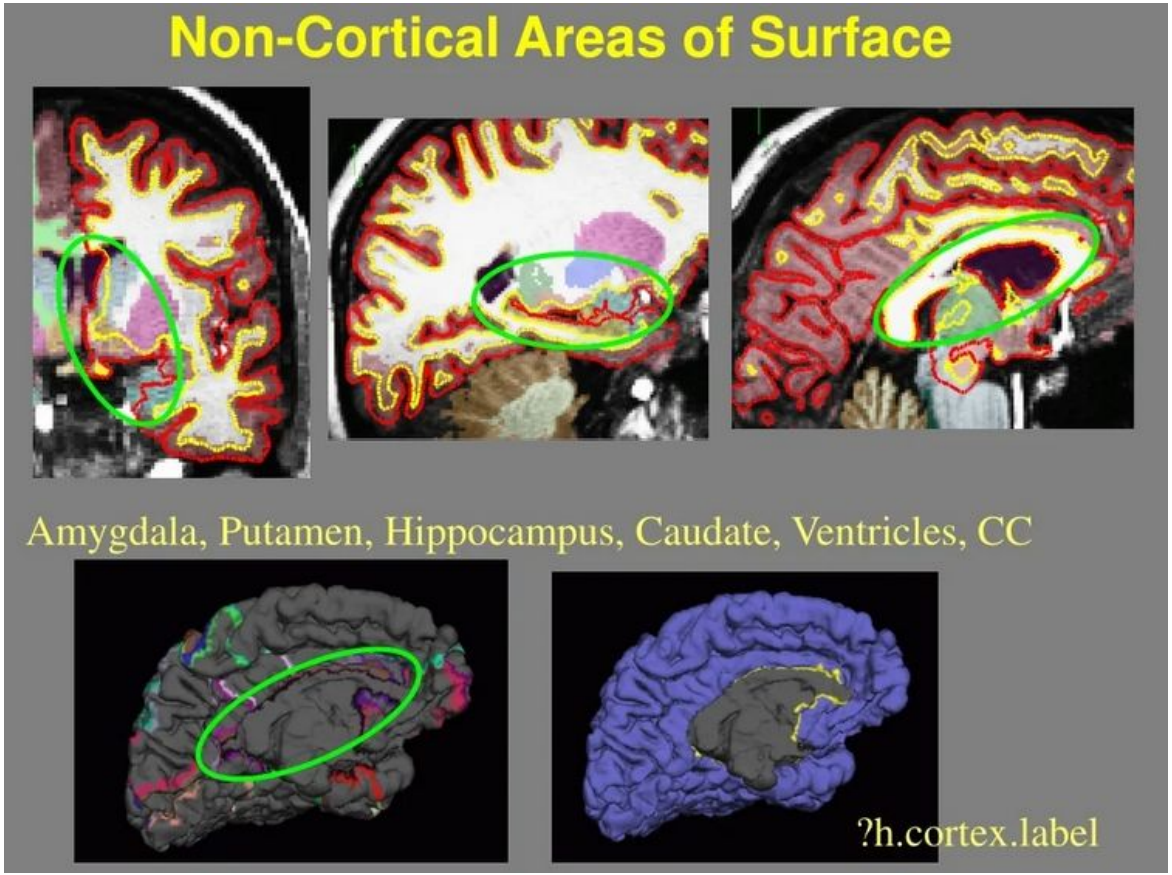
Cortical thickness



- Along normal vector of white/gray surface
- 1-5mm thick

Context: Morphometric measures of interest

Cortical surface area



Cortical surface Area (mm^2)

Study question

- How **input MRI image contrast(s) (added to T1w)** and **manual editing procedures** affect the accuracy of **Freesurfer morphometrics** ?
  → Improving reproducibility (lack of standardized practise)
  → True neurobiological variation VS processing bias (lack of studies)

- Comparison:
  - T1w-MPRAGE
  - T1w-MPRAGE + T2w-FLAIR
  - T1w-MPRAGE + T2w-SPACE
  - w/wo manual intervention

→ 2 datasets

Carotid and Mind (CAM) primary cohort

Inclusion criterion:

- 50-85 yo
- varying degrees of vascular risk factors
  - w/wo extracranial carotid artery disease (ECAD)

Rejection criterion:

- dementia, major depression, life-limiting cancer, a history of transient ischemic attack (TIA) or stroke within the preceding six months

Post-imaging rejection cases:

- 17 cortical interacts, 4 wo T2-FLAIR, 10 severe motions

→ N = 123 subjects

	Characteristics
Characteristic	**N = 123**
Mean Age	72 (±7)
Sex	
Female	61 (50%)
Male	62 (50%)
Ethnicity	
Hispanic/Latino	15 (12%)
Not Hispanic/Latino	103 (84%)
Not Reported	2 (1.6%)
Unknown	3 (2.4%)
Education Years	
≤ 12	31 (25%)
13-16	60 (49%)
≥ 17	31 (25%)
Race	
White	117 (95%)
Asian	4 (3.3%)
Black	1 (0.8%)
Other	1 (0.8%)
Hypertension	80 (65%)
Hyperlipidemia	94 (76%)
Diabetes	38 (31%)
Smoking	64 (52%)
History of Stroke	11 (8.9%)
Carotid Stenosis	67 (55%)
Coronary Artery Disease	43 (35%)
Peripheral Artery Disease	22 (18%)

External ADNI cohort

Alzheimer's Disease Neuroimaging Initiative
2003, Michael W. Weiner

Inclusion criterion:

- 55-85 yo
- mild cognitive impairment (MCI) or cognitively normal (CN)
- Sagittal T1w MPRAGE, T2w FLAIR, and T2w SPACE acquired on a Siemens 3T system during the same first visit

	Characteristics
Characteristic	**N = 143**
Age, Mean (SD)	68 (7)
Sex,	
F	82 (57%)
M	61 (43%)
Race,	
Asian	5 (3.5%)
Black / African American	66 (46%)
More than one race	5 (3.5%)
Unknown	1 (0.7%)
White	66 (46%)
Ethnicity,	
Hispanic/Latino	19 (13%)
Not Hispanic/Latino	124 (87%)
Group,	
CN	59 (41%)
MCI	84 (59%)

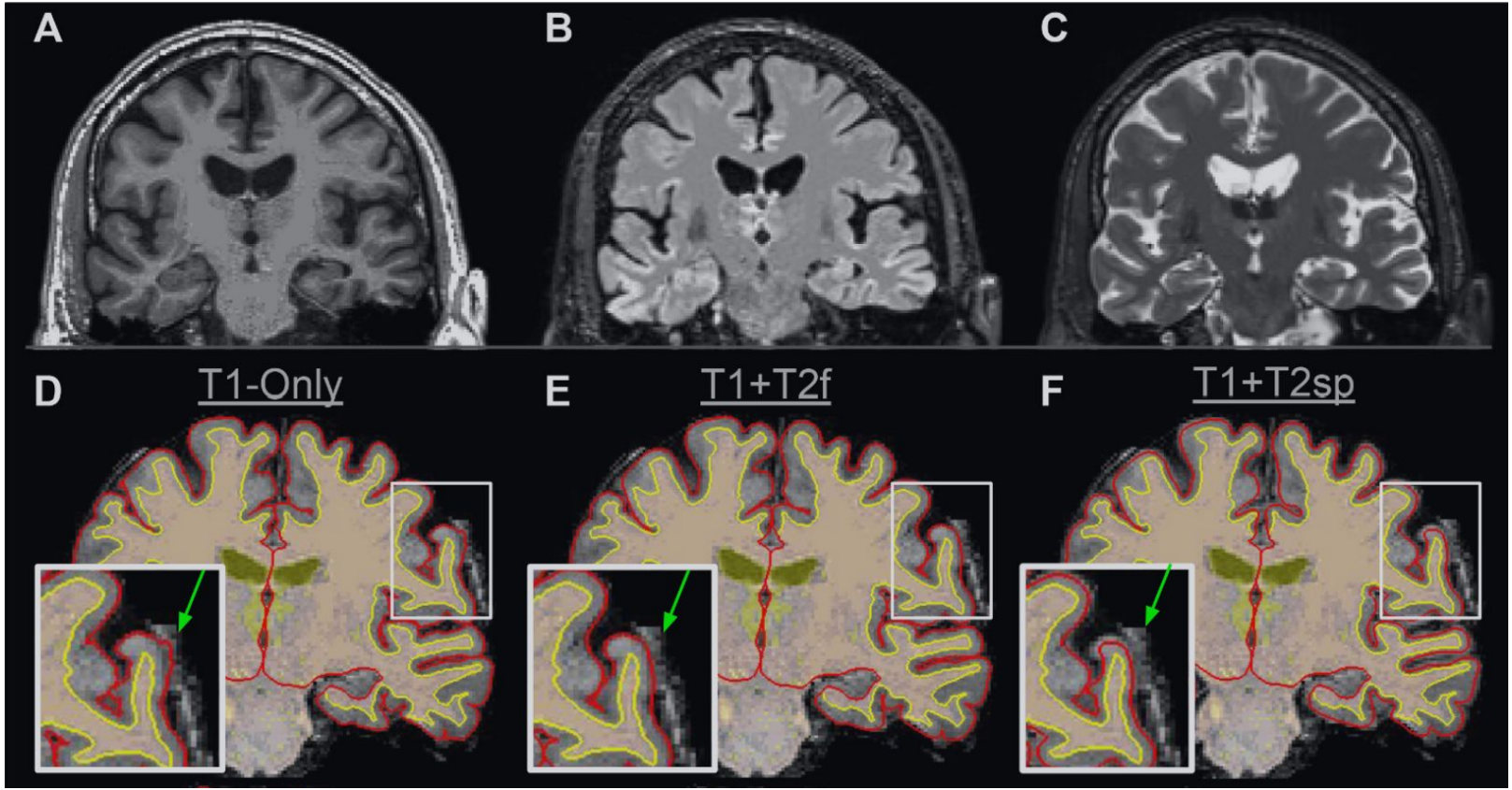
Carotid and Mind (CAM) secondary cohort

- 30 randomly selected combinations (T1, T1+T2f, T1+T2sp)
- one single experienced rater (HW) for manual corrections:
  - WM labels in wm.mgz
    - edit using brunch size 1 in coronal view
    - max 30 control points
    - only for ≥3 slices thick discrepancies
  - Pial surface adjustments to remove dura/venous sinuses

	Characteristics
Characteristic	**N = 30**
Age, Mean (SD)	71 (7)
Sex,	
Female	17 (57%)
Male	13 (43%)
Ethnicity,	
Hispanic/Latino	2 (6.7%)
Not Hispanic/Latino	27 (90%)
Not Reported	0 (0%)
Unknown	1 (3.3%)
Education Years,	
≤ 12	14 (47%)
13-16	9 (30%)
≥ 17	7 (23%)
Race,	
White	29 (97%)
Asian	0 (0%)
Black	1 (3.3%)
Other	0 (0%)
Hypertension,	24 (80%)
Hyperlipidemia,	24 (80%)
Diabetes,	5 (17%)
Smoking,	13 (43%)
History of Stroke,	2 (6.7%)
Carotid Stenosis %,	
≤ 29	13 (43%)
30-49	6 (20%)
50-69	5 (17%)
≥ 70	6 (20%)
Coronary Artery Disease,	11 (37%)
Peripheral Artery Disease,	4 (13%)

Results: quality check

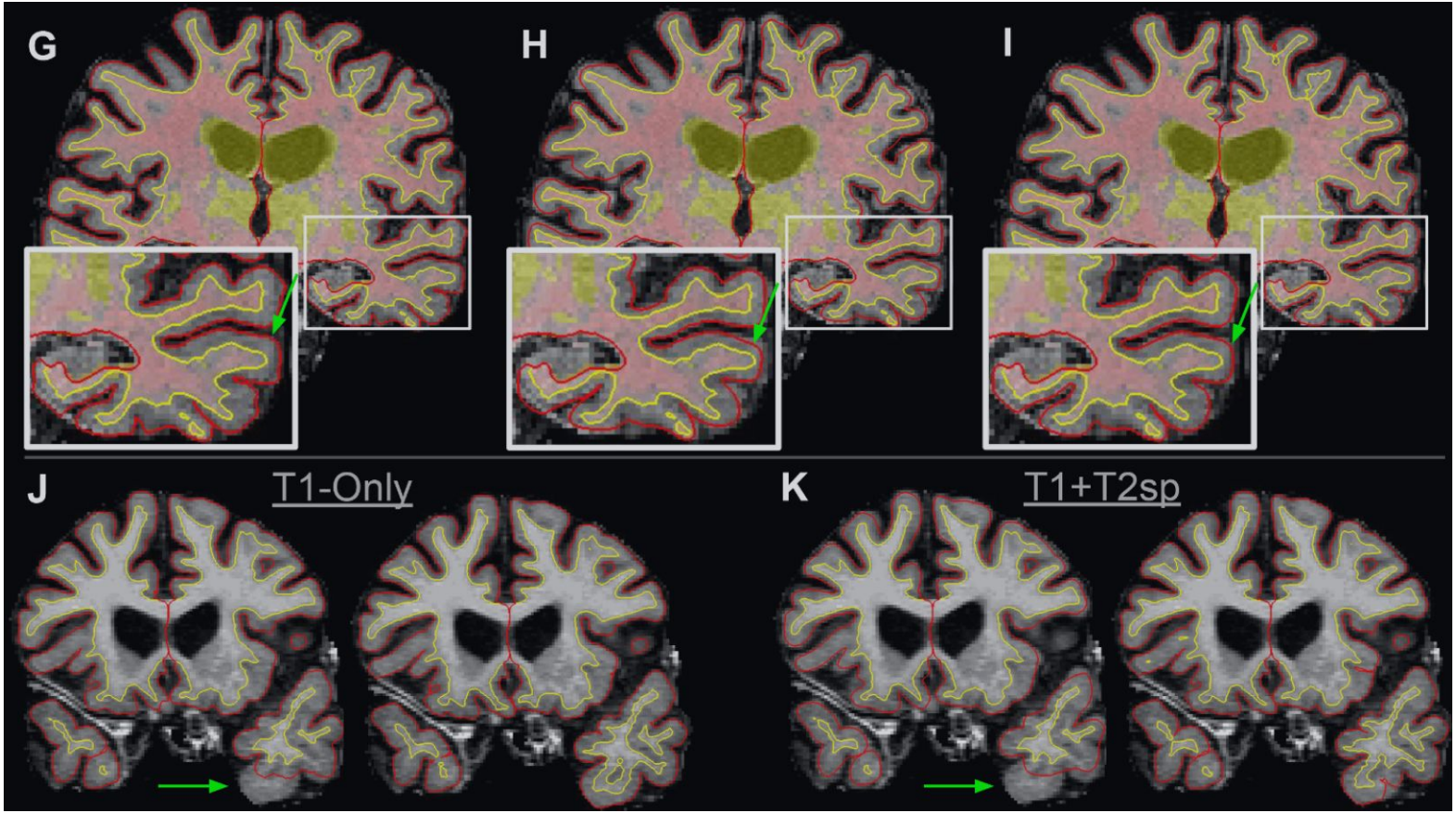
Choice of input contrast(s) affect automated segmentation (fig. 2)



FreeSurfer output surfaces overlayed on brainmask.mgz
T1-Only overestimates GM
T1+T2sp underestimates GM

Results: quality check

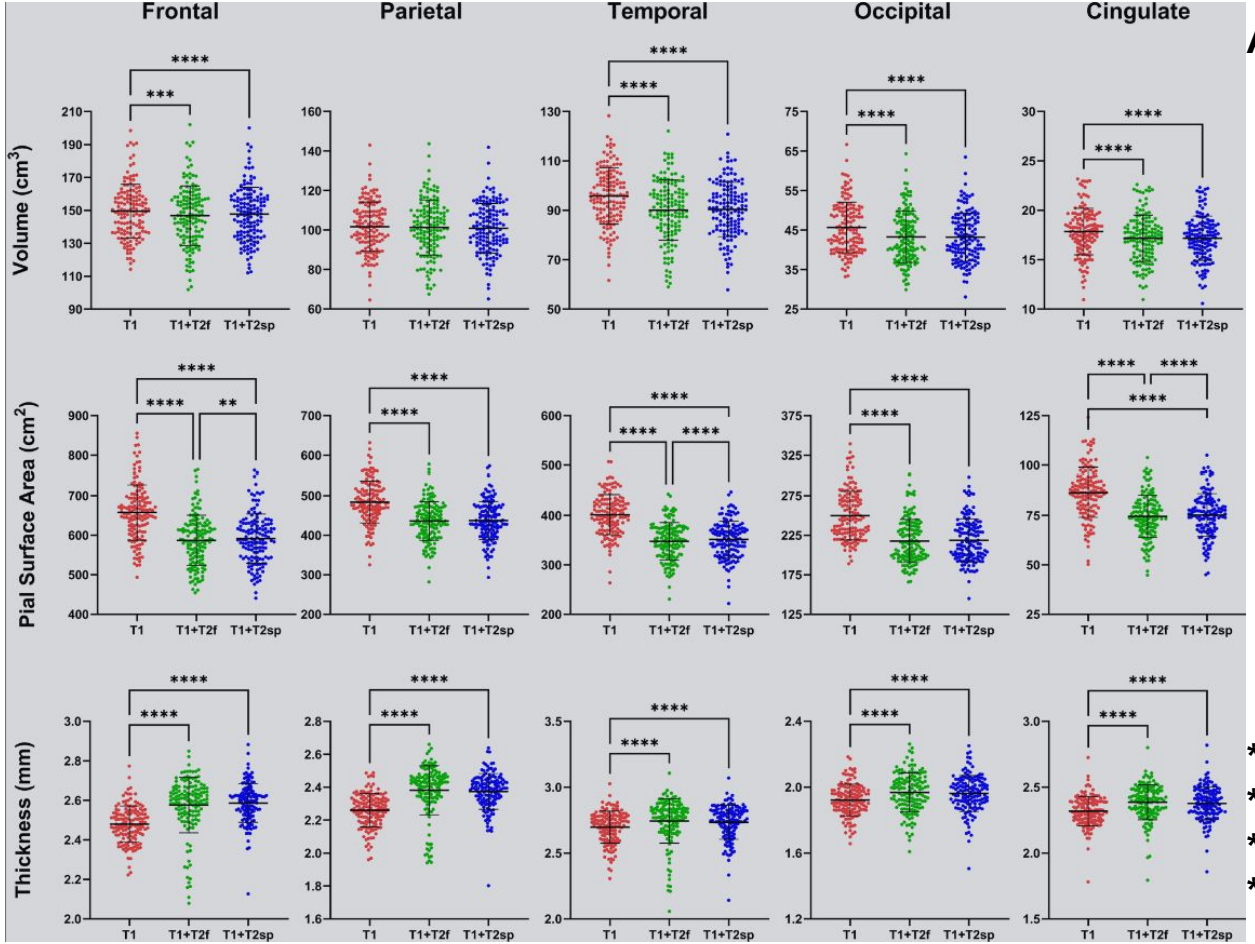
Choice of input contrast(s) affect automated segmentation (fig. 2)



T1-Only overestimates GM
T1+T2f underestimates GM

Pre-edit (left) underestimates temporal lobe

Results: ADNI

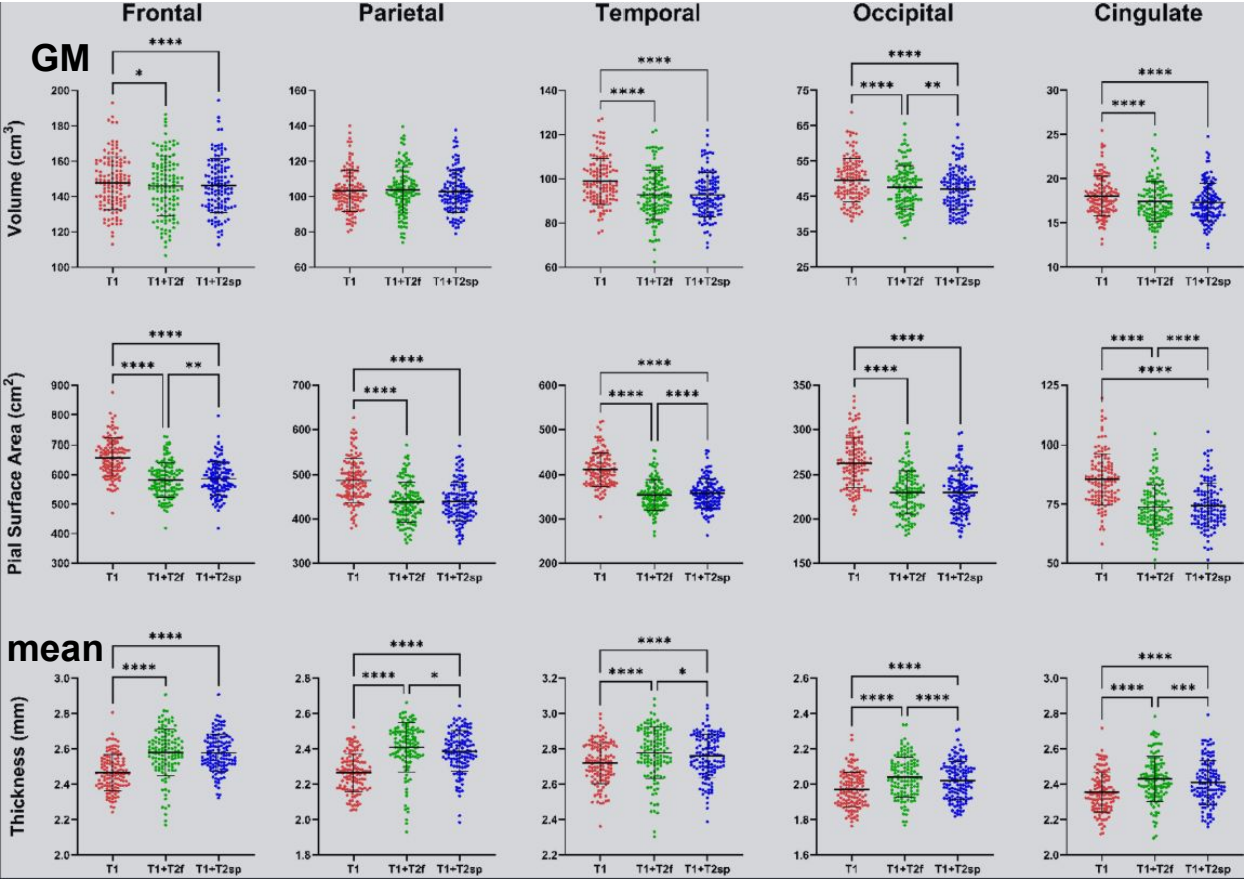


ADNI cohort (N = 143)

- **ANOVA results showed that scan type significantly affected morphometric estimates**
- **Tukey's post-hoc analysis (brackets) indicating that T1-only was significantly different from either other input combination in all regions and metrics except parietal volume ($p < 0.05$).**
- **WM surface area: no measurable differences (not displayed)**

*** $p < 0.05$**
**** $p < 0.01$**
***** $p < 0.001$**
****** $p < 0.0001$**

Results: pre-edit, CAN primary

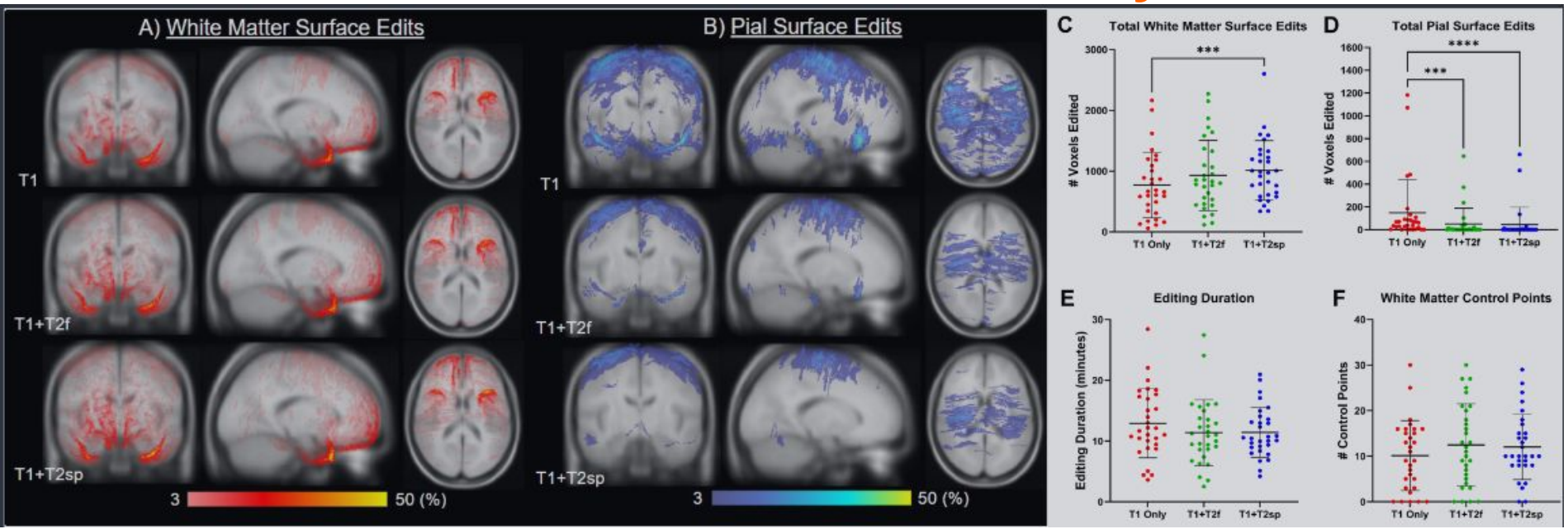


CAN primary cohort (N = 123)

- ANOVA results showed that scan type significantly affected morphometric estimates
- Tukey's post-hoc analysis (brackets) indicating that T1-only was significantly different from either other input combination in all regions and metrics except parietal volume ($p < 0.05$)
- WM surface area: no measurable differences (not displayed)

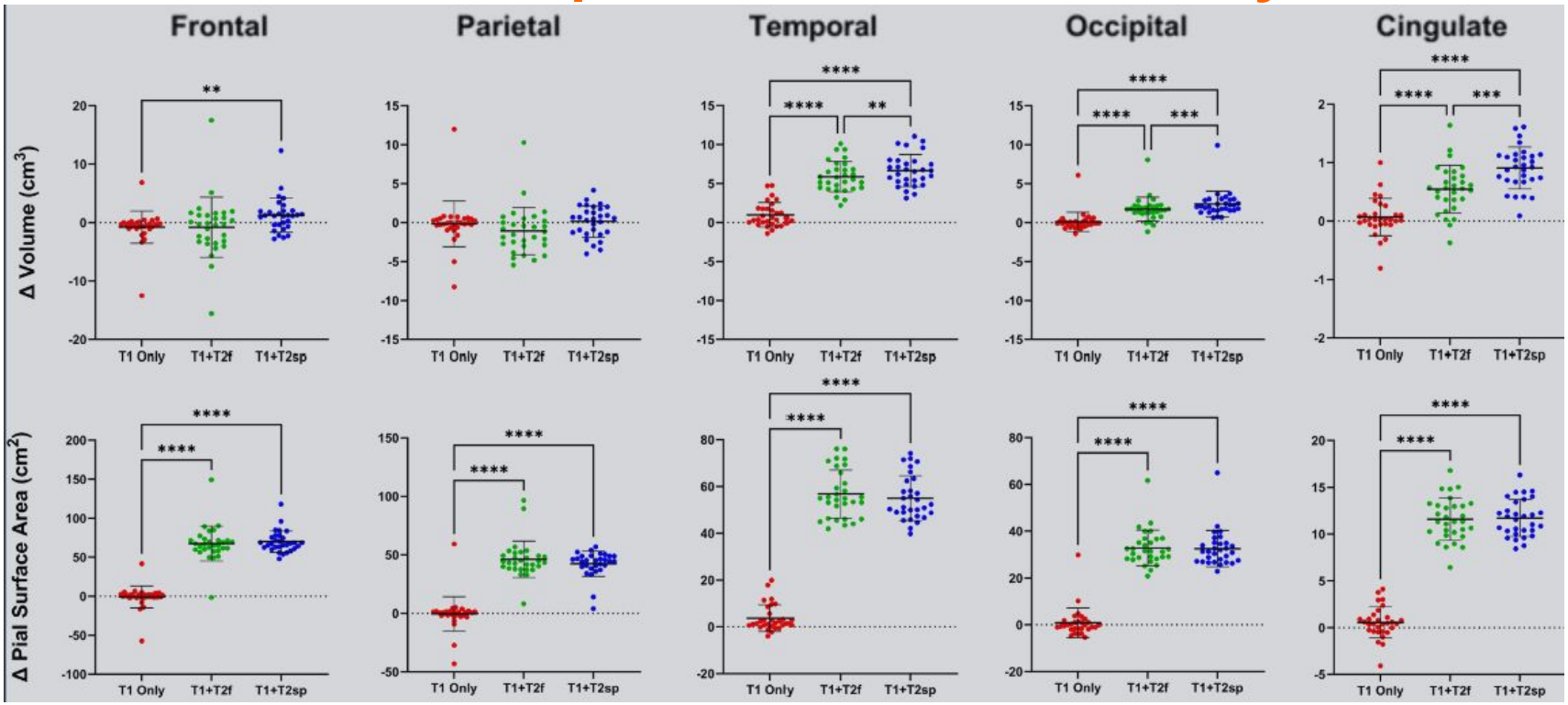
* $p < 0.05$
** $p < 0.01$
*** $p < 0.001$
**** $p < 0.0001$

Results: edit, CAN secondary



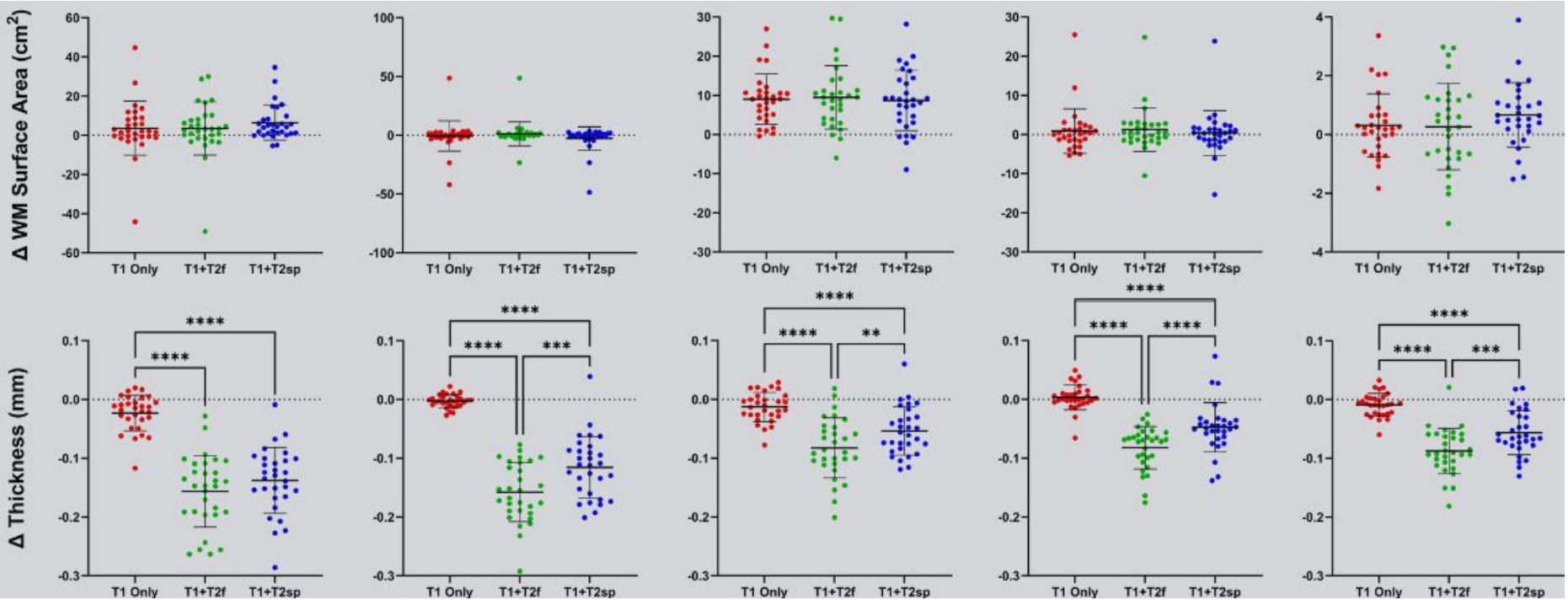
A) Maximum intensity projection maps of WM voxel edit frequency from the three input combinations. The color bar represents the percent of Secondary CAM cohort subjects who received WM surface edits per voxel. B) Maximum intensity projection maps depicting pial edit frequency where the color bar represents the percent of Secondary CAM cohort subjects who received pial surface edits per voxel. C) **T1+T2sp** had greater frequency of **WM surface edits** compared to T1-only ($p < 0.001$), primarily in regions of the medial temporal lobe. D) **T1-Only** derived outputs exhibited greater frequency of **pial surface edits** compared to T1+T2f and T1+T2sp, primarily in the frontal and parietal lobes. E) **Editing duration** was **not significantly different** across input scan type. F) The number of **WM control points** applied during the editing procedure also **did not differ significantly**.

Results: post-edit, CAN secondary



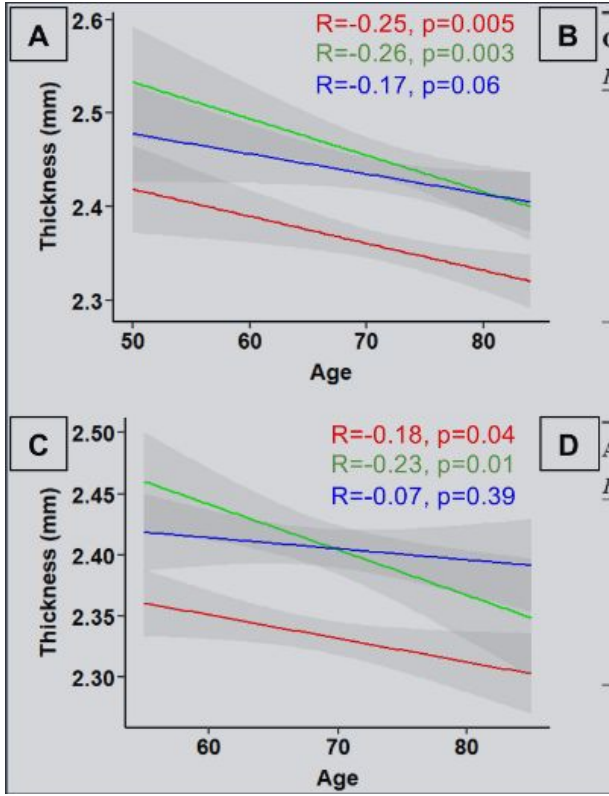
- Edits applied to outcomes from T2w input types resulted in significantly greater change in pial SA across all lobes.

Results: post-edit, CAN secondary



- No significant difference in lobe-wise change was observed across input scan type for WM surface area estimates.
- Substantial reductions in T1+T2-derived thickness and minimal changes in T1-only

Results: post-edit, CAN secondary



B

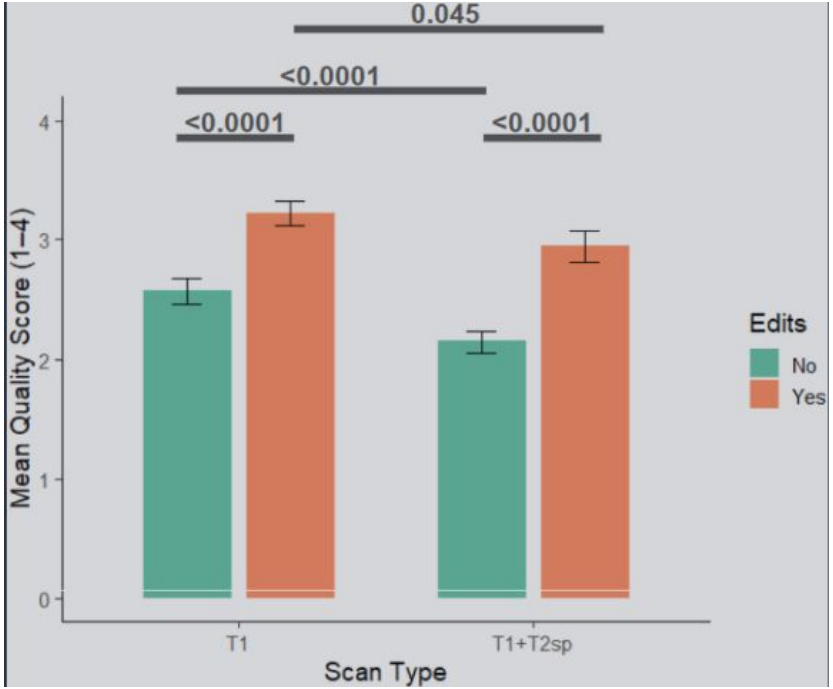
CAM Cohort	Thickness: T1-only			T1+T2f			T1+T2sp		
Predictors	*Estimates*	*CI*	*p*	*Estimates*	*CI*	*p*	*Estimates*	*CI*	*p*
Age	-0.23	-0.40 – -0.05	**0.011**	-0.25	-0.43 – -0.08	**0.005**	-0.14	-0.32 – 0.03	0.109
Sex [Male]	-0.25	-0.59 – 0.09	0.147	-0.23	-0.58 – 0.12	0.192	-0.20	-0.55 – 0.15	0.255
Race [Black/Other]	-0.30	-1.10 – 0.49	0.448	-0.28	-1.09 – 0.52	0.484	-0.46	-1.27 – 0.34	0.256
Ethnicity [Hispanic/Latino]	-0.39	-0.92 – 0.15	0.156	-0.12	-0.66 – 0.43	0.666	-0.49	-1.04 – 0.05	0.075
Smoking	-0.44	-0.80 – -0.09	**0.014**	-0.32	-0.68 – 0.04	0.078	-0.47	-0.82 – -0.11	**0.011**
Observations	123			123			123		
R^2 / R^2 adjusted	0.137 / 0.100			0.113 / 0.075			0.111 / 0.073		

D

ADNI Cohort	Thickness: T1-only			T1+T2f			T1+T2sp		
Predictors	*Estimates*	*CI*	*p*	*Estimates*	*CI*	*p*	*Estimates*	*CI*	*p*
Age	-0.24	-0.43 – -0.06	**0.009**	-0.25	-0.43 – -0.07	**0.007**	-0.10	-0.28 – 0.09	0.303
Sex [Male]	-0.22	-0.56 – 0.11	0.191	-0.18	-0.50 – 0.15	0.293	-0.37	-0.71 – -0.03	**0.034**
Race [Black/Other]	-0.46	-0.82 – -0.10	**0.013**	-0.25	-0.61 – 0.11	0.175	-0.32	-0.68 – 0.05	0.094
Ethnicity [Hispanic/Latino]	-0.20	-0.69 – 0.28	0.411	-0.65	-1.12 – -0.17	**0.008**	-0.07	-0.56 – 0.42	0.773
Observations	143			143			143		
R^2 / R^2 adjusted	0.083 / 0.057			0.109 / 0.083			0.054 / 0.027		

(A/C) Linear regressions illustrate whole-brain mean cortical thickness versus age for CAN/ADNI
(B/D) Multiple linear regression models adjusting for age, sex, race, ethnicity, and smoking for CAN/ADNI

Results: post-edit, CAN secondary



- Manual editing significantly improves segmentation quality (1 = poor, 4 = excellent)
- Prior to editing only, T1-only reconstructions had a significantly higher quality score (mean score: 2.57 ± 0.60) than T1+T2sp (mean score: 2.14 ± 0.49).

Conclusion

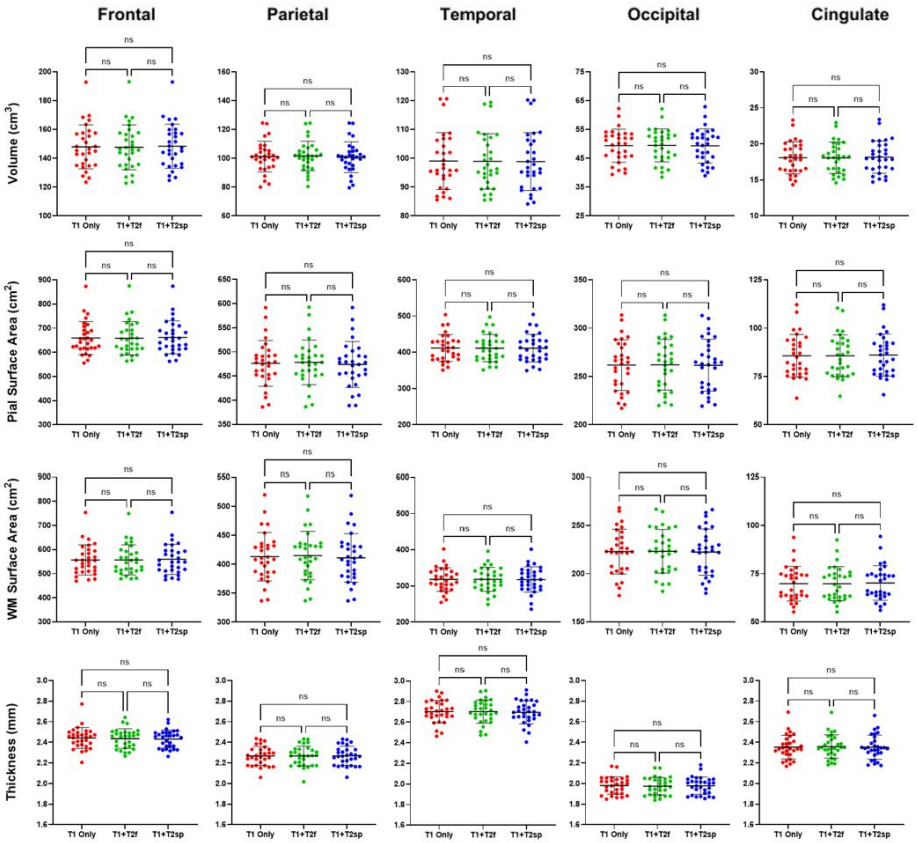
- Freesurfer limits: pre-edit segmentation **quality scores <3 on average**
- **T1+T2f and T1+T2sp** segmentations yielded significantly **lower cortical volume and pial surface area**, and greater cortical thickness estimates compared to T1-only
  - Largest size effects: +14 % (58 cm2) for pial surface area and +6 % (or 6 cm3) for volume in the temporal lobe
- **Difference between T1+T2f and T1+T2sp was generally small** (< 1.08 %) across lobes and measures
  - suggesting some degree of systematic bias in processing
- **T1-only reconstructions provide the most reliable out-of-box estimates**, and the addition of a T2w contrast can produce significant deviations
  Both for CAN and ADNI: may suggest intrinsic to sequence effects:
  - T2sp: high frequency signal contributions broadening the point spread function
  - T2-FLAIR: imperfect CSF suppression, larger slice thickness, and partial volume effects
- **Temporal lobe**: dura/vascular structures and susceptibility to B1 inhomogeneity can promote boundary misclassification, while T2w images improve pial boundary delineation, but negative improvement overall
- Significant associations between **cortical thickness** and **both age and smoking**

→ **Don’t age and don’t smoke !**

Thanks for your attention !

BONUS

Results: post-edit

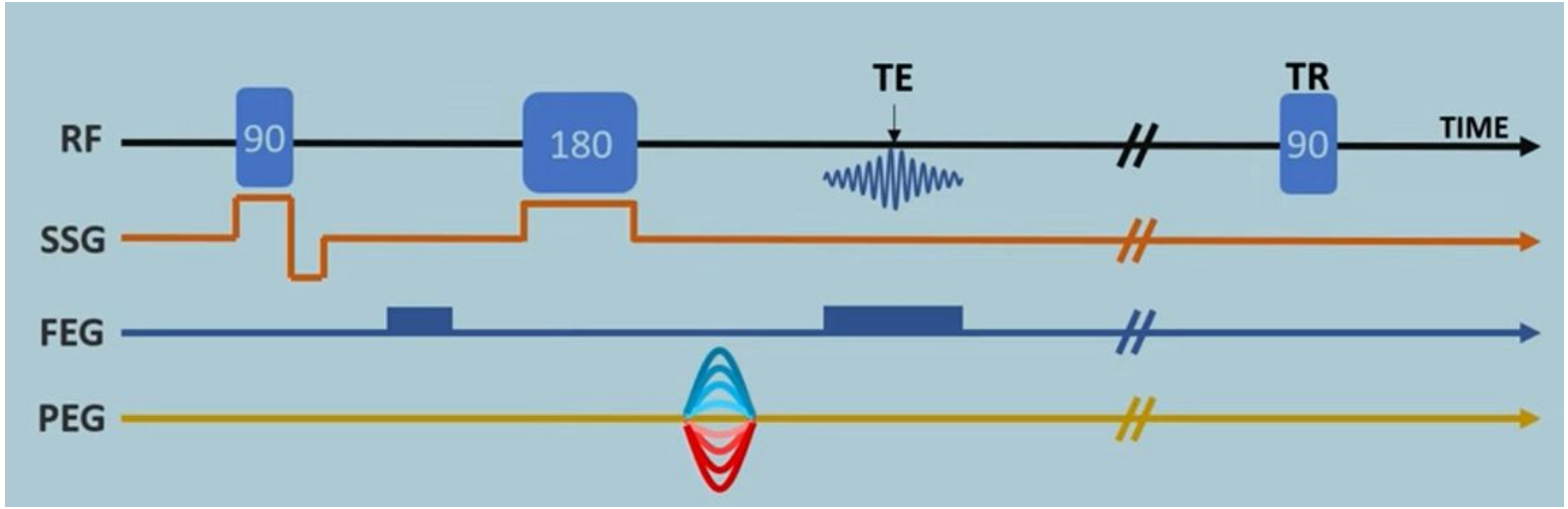


Supplemental Figure 3. Manual editing makes morphometric outcomes comparable across scan type. Edited T1-only, T1+T2f, and T1+T2sp outcomes for the five regions (Frontal, Parietal, Temporal, Occipital, Cingulate) are displayed: cortical volume (cm^3), pial surface area (cm^2), white-matter surface area (cm^2), and cortical thickness (mm). Points are individual subjects linked by lines (within-subject change). Brackets denote paired tests. No significant differences in morphometric estimates were observed after manual editing.

T1w MPRAGE

MPRAGE = Magnetization Prepared Rapid Gradient Echo

(TR/TE = 2300/2.98 ms, TI = 900 ms, flip angle = 9 degrees, voxel size = 1 mm isotropic, matrix = 240 × 256, 208 slices per slab, acquisition time = 5 min 12 s, bandwidth = 240 Hz/Px, in-plane parallel imaging using GRAPPA, acceleration factor = 2



https://www.youtube.com/watch?v=lln6_J3V514

https://www.youtube.com/watch?v=TQegSF4ZilQ

T2w FLAIR

- FLAIR (Fluid-Attenuated Inversion Recovery)

TR/TE = 4800/441 ms, TI = 1650 ms, echo spacing = 3.42 ms, ETL = 243, voxel size = 1 × 1 × 1.2 mm3, matrix = 256 × 256, 160 slices per slab, acquisition time = 5 min 28 s, bandwidth = 849 Hz/Px, GRAPPA acceleration factor = 3

T2w SPACE

- SPACE (Sampling Perfection with Application optimized Contrasts using different flip angle Evolution)

TR/TE = 3200/564 ms, echo spacing = 3.46 ms, ETL = 314, voxel size = 1 mm isotropic, matrix size = 240 × 256, FOV = 240 × 256 mm, 208 slices per slab, acquisition time = 4 min 53 s, bandwidth = 751 Hz/ Px, GRAPPA acceleration factor = 2

Feature	T2 FLAIR	T2 SPACE
Type	T2 + inversion recovery	3D T2 turbo spin echo
Fluid (CSF)	❌ Dark (suppressed)	✅ Bright
Purpose	Highlight lesions near fluid	High-resolution anatomy
Dimension	Usually 2D (or 3D variants)	Fully 3D
Strength	Lesion detection	Structural detail